Proprietary to General Electric Company
Direction 5438996-8EN, **Rev#** 9
Brivo CT385 Series Service Methods (Adv)
Module Version 4.0, **Version Date** 1/9/14

Quality Assurance Test

(Click Here for Summary Version)

1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	30 mins	Not Applicable

2 Overview

This document provides the necessary steps to ensure system image quality before releasing.

Last revised:	09 January 2014

3 Preliminary Requirements

3.1 Tools and Test Equipment

Item	Qty	Effectivity	Part#	Manufacturer
Phantom Holder	1	-	2331933-2	-
Quality Assurance Phantom	1	-	5498268	-
Quality Assurance Phantom	1	-	2144715	-
Bubble Level	1	-	-	-

3.2 Required Conditions

Condition	Reference	Effectivity
Recommend procedure completion by trained personnel.	-	-

4 Procedure

4.1 Phantom Setup

1. Record the phantom serial number on GE e4879 Component Replacement Form.
2. Perform *Center Phantom*

4.2 Acquire Scans

1. Select [NEW PATIENT] from bottom of *Scan* monitor.
2. Type the following entries in the listed *Patient Information* fields:
    a. Patient ID: **Service**
    b. Patient Name: **Quality Assurance Test**
3. Select *Anatomical Selector Service* tab from *Protocol Selection*.
4. Select [MANUFACTURING], [IMGSER 20CMQA] from *Anatomical Selector Service* tab.
5. Select [CONFIRM].
6. Press [START] on SCIM / GSCB when it flashes. Scans are performed.

7. Select [NEXT SERIES].
8. Select [CONFIRM].
9. Press [MOVE TO SCAN] on SCIM /GSCB if it flashes.
10. Press [START] on SCIM / GSCB when it flashes. Scans are performed.
11. Repeat Steps [7] - [10] until all series have been performed and completed.
12. When all series are completed, select [END EXAM].
13. If a PPS Response window is displayed, select [COMPLETE].

4.3 Analysis

1. Select *SERVICE* desktop and move *SERVICE* desktop window to left monitor.
2. Select *IMAGE QUALITY* tab from *SERVICE* desktop.
3. Select *IMAGE ANALYSIS* and move *IMAGE ANALYSIS* window to left monitor.
4. Select scanned examination from right monitor List/Select browser.

   NOTE: Series 1 and Image 1 should be selected. If not, select them.

5. Press [AUTO 1X] radio button on *IMAGE ANALYSIS* window.
6. Select [AUTO 1X] selection button, [IMGSER 20QA], and first test choice.

   NOTE: Number of test choices correspond to the number of series in the scanned examination. The order of the test choice correlates with the series number of the scanned examination (Series 1/Test 1, Series 2/Test 2, etc.).

7. Select [ACCEPT].

   NOTE: At test completion, test data results, including specification limits and Pass/Fail results, are displayed.

   NOTE: At the top of the results window, the test Pass/Fail summary result is posted.

8. Record results on *GE e4879 Component Replacement Form*.
9. Click [DISMISS] from *Report* window.
10. Select next series from right monitor List/Select browser.
11. From *IMAGE ANALYSIS* window, select [AUTO 1X] selection button, and select [IMGSER 20QA] and next test choice.

    NOTE: If test choice has multiple selections, execute each selection.

12. Select [ACCEPT].

    NOTE: If test requires user input, follow instructions in *IMAGE ANALYSIS* window lower right pane.

    NOTE: At test completion, test data results, including specification limits and Pass/Fail results, are displayed.

    NOTE: At the top of the results window, the test Pass/Fail summary result is posted.

13. Record results on the *GE e4879 Component Replacement Form*.
14. Click [DISMISS] from *Report* window.
15. Repeat steps [10] - [14] until all tests have been performed.
16. When all tests are completed, click [QUIT].

    NOTE: If any test fails, repeat failed test. If the test fails persistently, perform appropriate corrective action.

4.4 Image Review for Bands and Streaks

1. From *IMAGEWORKS* desktop List/Select browser, select scanned examination, and Series 3.
2. Select [VIEWER] and review all images from series for band and streak artifacts.
3. If any images depict bands, perform steps in Band Analysis, *Band Analysis,* on those images.
4. If any images depict streaks, perform *Streak Test*.
5. In viewer, select series 5 and review all images from series for band and streak artifacts.
6. If any images depict bands or streaks, repeat Step [3] or [4].
7. Repeat Steps [5] - [6] for Series 7 and 10.

4.5 Band Analysis

1. Select *SERVICE* desktop and move *SERVICE* desktop window to left monitor.
2. Select *IMAGE QUALITY* tab from *SERVICE* desktop.
3. Select [IMAGE ANALYSIS] and move *IMAGE ANALYSIS* window to left monitor.
4. From the List/Select browser on right monitor, select image to be analyzed.
5. Press [MANUAL] radio button on the *IMAGE ANALYSIS* window.
6. Click [MANUAL] selection button, then [BAND].
7. Click [ACCEPT]. The image is displayed in *IMAGE ANALYSIS* window with a Band ROI.
8. Follow instructions, in *IMAGE ANALYSIS* window lower right pane, to size Band ROI and analyze.
9. From results, if Avg. Ref. Band Factor is out of specification range of +/-2.8, perform appropriate corrective action.
10. When all tests are completed, click [QUIT].

5 Finalization

Store the *GE e4879 Component Replacement Form* with system history records. The results are stored in a text file on the scanner and can be retrieved at `/usr/g/service/state/image_analysis.log`.

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